Discrete Mathematics

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SD - MATH112

February 8, 2026

Propositions

Definition

A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

Examples:

- Washington, D.C., is the capital of the United States of America.
- Toronto is the capital of Canada.
- $1 + 1 = 2$.
- $2 + 2 = 3$.
- What time is it?
- Read this carefully.
- $x + 1 = 2$.
- $x + y = z$.

Propositions

We use letters to denote propositional variables (or sentential variables), that is, variables that represent propositions, just as letters are used to denote numerical variables. The conventional letters used for propositional variables are $p, q, r, s, \ldots$.
The truth value of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition.
Propositions that cannot be expressed in terms of simpler propositions are called atomic propositions.
The area of logic that deals with propositions is called the **propositional calculus** or **propositional logic**. It was first developed systematically by the Greek philosopher Aristotle more than 2300 years ago.
New propositions, called **compound propositions**, are formed from existing propositions using **logical operators**.

Negation

Definition

Let p be a proposition. The negation of p, denoted by $\neg p$ (also denoted by $\overline{p}$), is the statement "It is not the case that p." The proposition $\neg p$ is read "not p." The truth value of the negation of p, $\neg p$, is the opposite of the truth value of p.

Examples

- If p= "Michael's PC runs Linux", then $\neg p$= "Michael's PC does not run Linux"
- If q= "Vandana's smartphone has at least 32 GB of memory", then $\neg q$= "Vandana's smartphone has less than 32 GB of memory".

p	$\neg p$
T	F
F	T

Conjunction

Definition

Let p and q be propositions. The conjunction of p and q, denoted by $p \wedge q$, is the proposition "p and q." The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

Definition

Let p and q be propositions. The disjunction of p and q, denoted by $p \vee q$, is the proposition "p or q." The conjunction $p \vee q$ is true when at least one of p and q is true and is false otherwise.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

XOR

Definition

Let p and q be propositions. The exclusive or of p and q, denoted by $p \oplus q$, is the proposition "p exlcusive or q." The conjunction $p \vee q$ is true when exactly one of p and q is true and is false otherwise.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Conditional Statements

Definition

Let p and q be propositions. The conditional statement $p \to q$ is the proposition "if p, then q." The conditional statement $p \to q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \to q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Conditional Statements

Example

- Let p be the statement "Maria learns discrete mathematics" and q the statement "Maria will find a good job." Express the statement $p \rightarrow q$ as a statement in English.
  Solution: If Maria learns discrete mathematics, then she will find a good job."
- What is the value of the variable x after the statement
  $$\text{if } 2 + 2 = 4 \text{ then } x := x + 1$$
  if $x = 0$ before this statement is encountered? (The symbol $:=$ stands for assignment. The statement $x := x + 1$ means the assignment of the value of $x + 1$ to x.)

Converse, Contrapositive, Inverse

Consider the conditional statement $p \to q$.

- Its **converse** is $q \to p$,
- Its **contrapositive** is $\neg q \to \neg p$,
- Its **inverse** is $\neg p \to \neg q$.

Example
Find the contrapositive, the converse, and the inverse of the conditional statement "The home team wins whenever it is raining." The statement is the same thing as "If it is rainig, then the home team wins".
Solution:
Converse: "If the home team wins, then it is raining".
Contrapositive: "If the home team does not win, then it is not raining."
Inverse: "If it is not raining, then the home team does not win."

Biconditionals

Definition

Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition "p if and only if q." The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Biconditionals

Example: Construct the truth table of the compound proposition

$$(p \vee \neg q) \rightarrow (p \wedge q).$$

Precedence of Logical Operations

operator	precedence
$\neg$	1
$\wedge, \vee$	2
$\rightarrow$	3
$\leftrightarrow$	4

Logic and Bit Operations

truth value	bit
T	1
F	0

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
1	1	1	1	0
1	0	1	0	1
0	1	1	0	1
0	0	0	0	0

Logic and Bit Operations

Definition

A bit string is a sequence of zero or more bits. The length of this string is the number of bits in the string.

Example. 101010011 is a bit string of length nine.

Example. Find the bitwise OR, bitwise AND, and bitwise XOR of the bit strings 01 1011 0110 and 11 0001 1101.

01 1011 0110 11 0001 1101	
11 1011 1111	bitwise OR
01 0001 0100	bitwise AND
10 1010 1011	bitwise XOR

Logical Equivalence

Definition

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a **tautology**. A compound proposition that is always false is called a **contradiction**. A compound proposition that is neither a tautology nor a con- tradiction is called a contingency.

Example

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Therefore, for every p, $p \vee \neg p$ is a tautology and $p \wedge \neg p$ is a contradiction.

Logical Equivalence

Definition

The compound propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

Example: (De Morgan's Laws)

- $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$
- $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$

Example: $p \rightarrow q$ and $(\neg p) \vee q$ are logically equivalent.
Example: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$. Similarly, we have $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

Logical Equivalence

Example: Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Satisfiability

Definition

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true. When no such assignments exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**.

When we find a particular assignment of truth values that makes a compound proposition true, we have shown that it is satisfiable; such an assignment is called a solution of this particular satisfiability problem. However, to show that a compound proposition is unsatisfiable, we need to show that every assignment of truth values to its variables makes it false.

Example: Determine whether each of the compound propositions $(p \vee \neg q) \wedge (q \wedge \neg r) \wedge (r \vee \neg p)$, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$, and $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable.

Quantifiers

Definition

The universal quantification of $P(x)$ is the statement

"$P(x)$ for all values of x in the domain."

We write $\forall x P(x)$. $\forall$ is called the universal quantifier. If x is an element such that $P(x)$ is false we say that x is a counterexample to $\forall x P(x)$.

Definition

The existential quantification of $P(x)$ is the statement

"There exists an element x in the domain such that $P(x)$."

We write $\exists x P(x)$. $\exists$ is called the existential quantifier.

Quantifiers

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for all x	$P(x)$ is false for some x.
$\exists x P(x)$	There is an x : $P(x)$ is true	$P(x)$ is false for every x

Example Let $P(x)$ be the statement "$x + 1 > x$". If the domain consists of all real numbers, then $\forall x P(x)$ is true.
Example Let $Q(x)$ be the statement $x < 2$. If the domain is $\mathbb{R}$, then $\forall x Q(x)$ is false.
Example Let $R(x)$ be the statement $x^2 > 0$. Then, if the domain is $\mathbb{R}$, then $\forall x R(x)$ is false. 0 is a counterexample to the statement $\forall x R(x)$.

Quantifiers

The truth value may depend on the domain!

Example What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?

When the domain is $\mathbb{R}$, .5 is a counterexample to $\forall x P(x)$. Hence $\forall x P(x)$ is false.

When the domain is $\mathbb{Z}$ then $\forall x P(x)$ is true. In order to emphasize the domain sometimes I write $\forall x \in \mathbb{Z} \;\; P(x)$.

Example Let $P(x)$ be the statement that $x > 3$. Then $\exists x P(x)$ is true if the domain is $\mathbb{R}$.

Example Let $Q(x)$ be the statemnt $x = x + 1$ (the domain is $\mathbb{R}$). Then $\exists x Q(x)$ is false.

Quantifiers

Uniqueness quantifier: $\exists!$.
$\exists! x P(x)$ means there is one and only one x such that $P(x)$.
Example: Let $P(x)$ be the statement $x^3 = 8$ (the domain is $\mathbb{R}$). Then $\exists! x P(x)$ is true.
Example: Let $Q(x)$ be the statement $x^2 = 1$ (the domain is $\mathbb{R}$). Then $\exists! x Q(x)$ is false. Because $-1 \neq 1$ and $(-1)^2 = 1^2 = 1$.

Quantifiers

Definition

Statements involving predicates and quantifiers are logically equivalent if and only if they have the same truth value no matter which predicates are substituted into these statements and which domain of discourse is used for the variables in these propositional functions. We use the notation $S \equiv T$ to indicate that two statements S and T involving predicates and quantifiers are logically equivalent.

Example The statements $\forall x(P(x) \wedge Q(x))$ and $\forall x P(x) \wedge \forall x Q(x)$ are logically equivalent...

Negating Quantified Expressions

Fact

$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$, and
$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$.

Example The negation of $\forall x(x^2 > x)$ is $\exists x(x^2 \leq x)$.
The negation of $\exists x(x^2 = 2)$ is $\forall x(x^2 \neq 2)$.
Example $\neg\forall x(P(x) \rightarrow Q(x))$ is logically equivalent to $\exists x(P(x) \wedge \neg Q(x))$.

Nested Quantifiers

Examples

- $\forall x \exists y (x + y = 0)$.
- $\forall x \forall y (x + y = y + x)$.
- $\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$.
- $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$! But, in general, the order of quantifiers is important!
- $\forall x \exists y (x + y = 0)$ is true for every value of x and y in the domain $\mathbb{R}$. But $\exists x \forall y (x + y = 0)$ is false.
- Let $Q(x, y, z)$ be the statement $x + y = z$ (domain is $\mathbb{R}$). Then $\forall x \forall y \exists z Q(x, y, z)$ is true and $\exists z \forall x \forall y Q(x, y, z)$ is false. Why?

Nested Quantifiers

- The statement "the sum of two positive integers is always positive" can be expressed as $\forall x \forall y((x > 0) \wedge (y > 0)) \rightarrow (x + y > 0)$. If we restrict the domain to the set of positive numbers, $\mathbb{R}_{>0}$, then it can be expressed simply by $\forall x \forall y(x + y > 0)$.
- In calculus, $\lim_{x \to a} f(x) = A$ means
$$\forall \varepsilon > 0 \exists \delta > 0 \forall x(0 < |x - a| < \delta) \rightarrow (|f(x) - A| < \varepsilon).$$
Similarly, f is continuous at a if
$$\forall \varepsilon > 0 \exists \delta > 0 \forall x(|x - a| < \delta) \rightarrow (|f(x) - f(a)| < \varepsilon).$$

Negating Nested Quantifiers

Example: Express the negation of the statement $\forall x \exists y(xy = 1)$ so that no negation precedes a quantifier.
$(\exists x \forall y)(xy \neq 1))$
Example Use quantifiers to express the statement that "There does not exist a woman who has taken a flight on every airline in the world." $(\neg \exists w \forall a(P(w, f) \wedge Q(f, a))$, where $P(w, f)$ is "w has taken f" and $Q(f, a)$ is "f is a flight on a".)
Example Use quantifiers and predicates to express the fact that $\lim_{x \to a} f(x)$ does not exist where $f(x)$ is a real-valued function of a real variable x and a belongs to the domain of f.

$$\forall L \exists \varepsilon > 0 \forall \delta > 0 \exists x[(0 < |x - a| < \delta) \wedge (|f(x) - L| \geq \varepsilon)].$$

More Examples

- $C(x)$ is "x has a computer" and
  $F(x, y)$ is "x and y are friends" the domain consists of students in the school. Then
  $$\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$$
  can be translated into English as "every student in the school has a computer or has a friend with a computer".
- The domain is again the set of students in the school $F(x, y)$ is as above. What is the meaning of

$$\exists x \forall y \forall z((F(x, y) \wedge F(x, z) \wedge (y \neq z) \rightarrow \neg F(y, z))?$$

Rules of Inference

Definition

An **argument** in propositional logic is a sequence of propositions. All but the final proposition in the argument are called **premises** and the final proposition is called the **conclusion**. An argument is valid if the truth of all its premises implies that the conclusion is true. An **argument form** in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is **valid** if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

From the definition of a valid argument form we see that the argument form with premises $p_1, p_2, \cdots, p_n$ and conclusion q is valid exactly when $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow q$ is a tautology.

Rules of Inference

The tautology $(p \wedge (p \rightarrow q)) \rightarrow q$ is the basis of the rule of inference called **modus ponens**, or the **law of detachment**.

$$\begin{array}{lll} p & & \\ p & \rightarrow & q \\ \hline \therefore & q & \end{array}$$

Example: "If $\sqrt{2} > \frac{3}{2}$, then $(\sqrt{2})^2 > \left(\frac{3}{2}\right)^2$". We know that $\sqrt{2} < \frac{3}{2}$. Hence we have $2 < \frac{9}{4}$.
The argument form is valid. But, one of the premises is false. Therefore we cannot conclude that the conclusion is true! Note that the conclusion is also false.

Rules of Inference

Rule of inference	Tautology	Name
p $p \to q$ $\therefore\ q$	$(p \wedge (p \to q)) \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore\ \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore\ p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore\ q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism

Rules of Inference

Rule of inference	Tautology	Name
p $\therefore \quad p \vee q$	$p \rightarrow (p \vee q)$	Addition
$p \wedge q$ $\therefore \quad p$	$(p \wedge q) \rightarrow p$	Simplification
p q $\therefore \quad p \wedge q$	$((p) \wedge (q)) \rightarrow (p \rightarrow q)$	Conjunction
$p \quad \vee \quad q$ $\neg p \quad \vee \quad r$ $\therefore \quad q \wedge r$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Rules of Inference

Example: State which rule of inference is the basis of the following argument: "It is below freezing and raining now. Therefore, it is below freezing now."

Example: State which rule of inference is used in the argument: If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, then we will have a barbecue tomorrow.

Example: Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

Examples: Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."

Rules of Inference

Computer programs have been developed to automate the task of reasoning and proving theorems. Many of these programs make use of a rule of inference known as **resolution**. This rule of inference is based on the tautology

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r).$$

The final disjunction in the resolution rule, $q \vee r$, is called the **resolvent**.

Example: Use resolution to show that the hypotheses "Jasmine is skiing or it is not snowing" and "It is snowing or Bart is playing hockey" imply that "Jasmine is skiing or Bart is playing hockey."

Remark: To construct proofs in propositional logic using resolution as the only rule of inference, the hypotheses and the conclusion must be expressed as **clauses**, where a clause is a disjunction of variables or negations of these variables. We can replace a statement in propositional logic that is not a clause by one or more equivalent statements that are clauses.

Rules of Inference

Example: Suppose we have a statement of the form $p \vee (q \wedge r)$. Because $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$, we can replace the single statement $p \vee (q \wedge r)$ by two statements $p \vee q$ and $p \vee r$, each of which is a clause.
We can also replace a statement of the form $\neg(p \vee q)$ by the two statements $\neg p$ and $\neg q$. (DeMorgan)
We can also replace a conditional statement $p \rightarrow q$ with the equivalent disjunction $\neg p \vee q$.
Example: Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.

Fallacies

The proposition $((p \to q) \wedge q) \to p$ is not a tautology, because it is false when p is false and q is true. However, there are many incorrect arguments that treat this as a tautology. In other words, they treat the argument with premises $p \to q$ and q and conclusion p as a valid argument form, which it is not. This type of incorrect reasoning is called the **fallacy of affirming the conclusion.**

Example: Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics.

Therefore, you did every problem in this book.

Fallacies

The proposition $((p \to q) \wedge \neg p) \to \neg q$ is not a tautology, because it is false when p is false and q is true. Many incorrect arguments use this incorrectly as a rule of inference. This type of incorrect reasoning is called the **fallacy of denying the hypothesis.**

Example:

Let p and q be as in the previous example. If the conditional statement $p \to q$ is true, and $\neg p$ is true, is it correct to conclude that $\neg q$ is true? In other words, is it correct to assume that you did not learn discrete mathematics if you did not do every problem in the book, assuming that if you do every problem in this book, then you will learn discrete mathematics?

Rules of Inference for Quantified Statements

$\forall x P(x)$ $\therefore P(c)$	Universal instantiation
$P(c)$ for any c $\therefore \forall x P(x)$	Universal generalization
$\exists x P(x)$ $\therefore P(c)$ for some element c	Existential instantiation
$P(c)$ for some element c $\therefore \exists x P(x)$	Existential generalization

Example: Show that the premises "Everyone in this discrete mathematics class has taken a course in computer science" and "Marla is a student in this class" imply the conclusion "Marla has taken a course in computer science."

Example: Show that the premises "A student in this class has not read the book," and "Everyone in this class passed the first exam" imply the conclusion "Someone who passed the first exam has not read the book."

Combining Rules of Inference

Because universal instantiation and modus ponens are used so often together, this combination of rules is sometimes called **universal modus ponens**.

$$\forall x(P(x) \rightarrow Q(x))$$
$$P(a) \text{ where } a \text{ is a particular element}$$
$$\therefore Q(a)$$

Example:
Assume that "For all positive integers n, if n is greater than 4, then n^2 is less than 2^n" is true. Use universal modus ponens to show that $100^2 < 2^{100}$.

Introduction to Proofs

- A **theorem** is a statement that can be shown to be true.
- Less important theorems sometimes are called **propositions**.
- We demonstrate that a theorem is true with a **proof**. A proof is a valid argument that establishes the truth of a theorem.
- The statements used in a proof can include **axioms** (or **postulates**), which are statements we assume to be true (for example, the axioms for the real numbers and the axioms of plane geometry), the premises, if any, of the theorem, and previously proven theorems.
- A less important theorem that is helpful in the proof of other results is called a **lemma**.
- A **corollary** is a theorem that can be established directly from a theorem that has been proved.
- A **conjecture** is a statement that is being proposed to be a true statement.

Direct Proofs

A **direct proof** of a conditional statement $p \to q$ is constructed when the first step is the assumption that p is true; subsequent steps are constructed using rules of inference, with the final step showing that q must also be true.

Definition

The integer n is *even* if there exists an integer k such that $n = 2k$, and n is *odd* if there exists an integer k such that $n = 2k + 1$. (Note that every integer is either even or odd, and no integer is both even and odd.) Two integers have the *same parity* when both are even or both are odd; they have *opposite parity* when one is even and the other is odd.

Example: Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."
Example: Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square. (An integer a is a **perfect square** if there is an integer b such that $a = b^2$.)

Proof by Contraposition

An extremely useful type of indirect proof is known as **proof by contraposition**.
The idea is to replace $p \to q$ by its equivalent $\neg q \to \neg p$.
Example: Prove that if n is an integer and $3n + 2$ is odd, then n is odd.
Example: Prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$

Vacuous Proofs

We can quickly prove that a conditional statement $p \rightarrow q$ is true when we know that p is false, because $p \rightarrow q$ must be true when p is false. Consequently, if we can show that p is false, then we have a proof, called a **vacuous proof**, of the conditional statement $p \rightarrow q$.

Example: Show that the proposition $P(0)$ is true, where $P(n)$ is "If $n > 1$, then $n^2 > n$" and the domain consists of all integers.

Trivial Proofs

We can also quickly prove a conditional statement $p \rightarrow q$ if we know that the conclusion q is true. By showing that q is true, it follows that $p \rightarrow q$ must also be true. A proof of $p \rightarrow q$ that uses the fact that q is true is called a **trivial proof**.
Example: Let $P(n)$ be "If a and b are positive integers with $a \geq b$, then $a^n \geq b^n$," where the domain consists of all nonnegative integers. Show that $P(0)$ is true.
Solution: Solution: The proposition $P(0)$ is "If $a \geq b$, then $a^0 \geq b^0$." Because $a^0 = b^0 = 1$, the conclusion of the conditional statement "If $a \geq b$, then $a^0 \geq b^0$" is true. Hence, this conditional statement, which is $P(0)$, is true. This is an example of a trivial proof. Note that the hypothesis, which is the statement "$a \geq b$," was not needed in this proof.

A Little Proof Strategy

First try to give a direct proof. If it seems to be difficult to give a direct proof, try to give a proof by contraposition.

Definition

The real number r is *rational* if there exist integers p and q with $q =\neq 0$ such that $r = p/q$. A real number that is not rational is called *irrational*. $\mathbb{Q}$ denotes the set of rational numbers.

Example: Prove that the sum of two rational numbers is rational. That is,

$$\forall r \forall s[(r \in \mathbb{Q}) \wedge (s \in \mathbb{Q})] \rightarrow (r + s) \in \mathbb{Q}.$$

Of course, the domain is the set $\mathbb{R}$.

Example: Prove that if n is an integer and n^2 is odd, then n is odd.

Proofs by Contradiction

Because the statement $r \wedge \neg r$ is a contradiction whenever r is a proposition, we can prove that p is true if we can show that $\neg p \rightarrow (r \wedge \neg r)$ is true for some proposition r. Proofs of this type are called **proofs by contradiction**.

Example: Show that at least four of any 22 days must fall on the same day of the week.

Example: Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Example: Give a proof by contradiction of the theorem "If $3n + 2$ is odd, then n is odd."

Proof of Equivalence

Example: Prove the theorem "If n is an integer, then n is odd if and only if n^2 is odd."
Example: Show that these statements about the integer n are equivalent:

p_1: n is even,

p_2: $n - 1$ is odd,

p_3: n^2 is even.

Counterexamples

To show that a statement of the form $\forall x P(x)$ is false, we need only find a **counterexample**, that is, an example x for which $P(x)$ is false.
Example: Show that the statement "Every positive integer is the sum of the squares of two integers" is false.

Mistakes in Proofs

Example: What is wrong with this famous supposed "proof" that $1 = 2$?
"Proof": We use these steps, where a and b are two equal positive integers.

1.	$a = b$	Given
2.	$a^2 = ab$	Multiply both sides of (1) by a
3.	$a^2 - b^2 = ab - b^2$	Subtract b^2 from both sides of (2)
4.	$(a - b)(a + b) = b(a - b)$	Factor both sides of (3)
5.	$a + b = b$	Divide both sides of (4) by $a - b$
6.	$2b = b$	Replace a by b in (5) because $a = b$ and simplify
7.	$2 = 1$	Divide both sides of (6) by b.

Mistakes in Proofs

Example: What is wrong with this "proof"?
"Theorem": If n^2 is positive, then n is positive.
"Proof ": Suppose that n^2 is positive. Because the conditional statement "If n is positive, then n^2 is positive" is true, we can conclude that n is positive.
Example: What is wrong with this "proof"?
"Theorem": If n is not positive, then n^2 is not positive.
"Proof ": Suppose that n is not positive. Because the conditional statement "If n is positive, then n^2 is positive" is true, we can conclude that n^2 is not positive.

Many incorrect arguments are based on a fallacy called **begging the question.** This fallacy occurs when one or more steps of a proof are based on the truth of the statement being proved. In other words, this fallacy arises when a statement is proved using itself, or a statement equivalent to it. That is why this fallacy is also called **circular reasoning**.

Example: Is the following argument correct? It supposedly shows that n is an even integer whenever n^2 is an even integer.

"Suppose that n^2 is even. Then $n^2 = 2k$ for some integer k. Let $n = 2l$ for some integer l. This shows that n is even."